HW2 QUESTION1 REPORT

In this question, I used **Template Method Design Pattern** becasue all three different genetic algorithm versions have very similar operations. Differences occurs only in selection and cross over types. The other operations like **generateInitialPopulation(),**

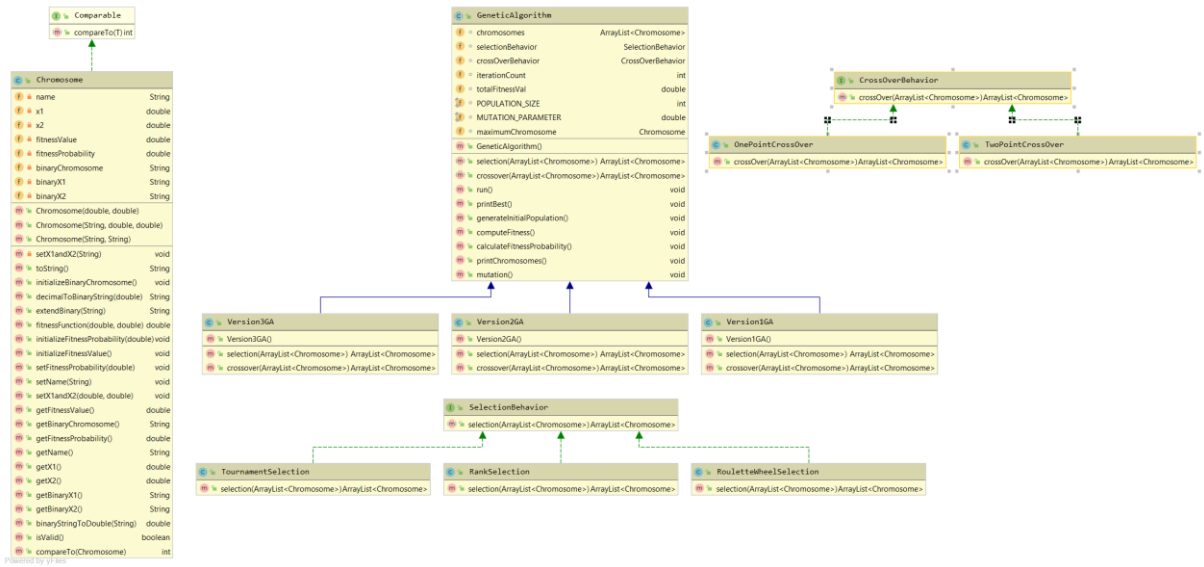
computeFitness(),mutation() have same implementation for three different genetic algorithm versions.

So I make abstract these **selection**() and **crossOver**() methods in abstract ***GeneticAlgorithm*** class. Then I create **Version1Ga**, **Version2GA** and **Version3GA** concrete classes. These concrete classes extends from abstract ***GeneticAlgorithm*** class and implements the **selection()** and **crossOver** methods according to their versions.

I create a **Chromosome** class for storing datas and operations about a cromosome.

Also I use Strategy Design pattern for Selection and Cross Over types to avoid the code duplication. I implements the Selection types by implementing the **SelectionBehavior** interface and I implements the Crossover types by implementing the **CrossOverBehavior** interface.

Class diagrams generated by Intellij IDE:



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